This article has been accepted for publication in IEEE Journal of Selected Topics in Signal Processing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JSTSP.2024.3386057

Model-Based Online Learning For Active ISAC Waveform Optimization

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***Abstract*—This paper proposes a Model-Based Online Learning (MBOL) framework for waveform optimization in integrated sensing and communications (ISAC) systems. In particular, the MBOL framework is proposed to enhance the ISAC performance under dynamic environmental conditions. Unlike Model-Free Online Learning (MFOL) methods, our approach leverages a rich structural knowledge of sensing, communications, and radio environments, offering better explainability and sample efficiency. This paper establishes a theoretical analysis of the proposed class of MBOL methods, showing essential performance conditions and convergence rates. This theoretical analysis is critical for understanding the potential of MBOL in active waveform optimization tasks. We demonstrate the proposed MBOL framework in multicarrier ISAC systems, focusing on the sub-carrier selection and power allocation problem. Via numerical experiments, we show that the proposed MBOL method outperforms the MFOL method in terms of sample efficiency. The results underline the potential of MBOL for improving the active waveform optimization performance in ISAC systems, particularly when sample efficiency and explainability are critical.**

***Index Terms*—joint radar-communications systems, reinforcement learning, online convex optimization, model-based learning, resource allocation, waveform optimization**

I. Introduction

Integrated sensing and communications (ISAC) systems aim to combine radio frequency (RF) sensing with communications [1]. This ability to provide joint sensing and communications services is considered a key new technology in emerging 6G wireless systems. ISAC systems reduce the overall system cost by using integrated hardware for sensing and communications tasks. Furthermore, ISAC systems mitigate the congestion in the radio spectrum by reusing and cooperatively sharing the radio spectrum among different tasks. Different models for ISAC systems have been proposed based on the level of cooperation and integration among the systems and sub-systems [2], [3]. Cooperative ISAC systems share information among the sensing and communications to use the resources more efficiently and manage mutual interference. This is in contrast to non-cooperative sensing and communications coexistence, where no information is shared among the nodes. In co-designed ISAC systems, the waveforms, the use of resources, and transceiver hardware for both sensing and communications tasks are designed jointly for mutual benefit.

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This paper has supplementary downloadable material available at http://ieeexplore.ieee.org/., provided by the author. The material includes supporting derivations and proofs for interested readers. Contact petteri.pulkkinen@aalto.fi for further questions about this work.

This paper considers the co-design and integration of radio frequency sensing and communications. Different names for such ISAC systems have been used in literature, such as joint radar-communication (JRC) [2] and dual-function radar communication (DFRC) systems [3]. Especially the DFRC system refers to a co-designed radar and communications system with fully integrated waveforms and hardware. Most current and emerging wireless systems use multi-carrier waveforms, such as 4G, 5G, and WiFi. Similarly, multi-carrier signals are used in active [4] and passive radars [5]. Such waveforms have many degrees of freedom, enabling adjusting and optimizing the use of radio resources in ISAC systems for the underlying radio environments to improve performance or to achieve certain operational requirements [6]. This paper develops online machine learning methods for waveform optimization and resource allocation problems in multi-carrier ISAC systems. The proposed learning framework is demonstrated in a sub-carrier selection and power allocation problem but also applies to other resources, such as antennas and beams of beamformers.

Traditionally, waveform optimization and resource allocation in ISAC systems have been tackled using batch-mode optimization methods, which do not incorporate learning processes. For example, in multicarrier ISAC systems, methods have been developed for optimizing waveforms to maximize the sensing performance while satisfying communications capacity constraints, or vice versa [7]–[12]. In [7]–[10], mutual information (MI) between the target scattering matrix and the received signal is used as a performance criterion for sensing. In contrast, Cramér–Rao bound (CRB) criterion for specific parameter estimation tasks are considered in [11], [12]. In distributed ISAC systems, an optimization method has been proposed in [13] to optimize power allocation across transmitting nodes by maximizing sum capacity while satisfying a localization accuracy constraint based on CRB. Another resource allocation method for distributed ISAC systems was proposed in [14] that optimizes transmit powers and dwell times for radar nodes as well as channel selections for communication nodes to minimize the Bayesian CRB while ensuring communications capacity constraints are satisfied.

The methods in [7]–[14], are based on scenarios where channel and interference statistics are known and wide-sense stationary (WSS), and the whole system is centrally optimized. However, this paper considers a dynamic shared spectrum scenario where the radio spectrum is time-frequency-space varying. In such environments, rapid changes in interference and channel conditions are often encountered. To address this, we study online learning methods specifically designed for optimizing waveforms in multicarrier ISAC systems under the dynamic conditions. *Online learning* in this context refers to

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machine learning techniques that enhance performance in real-time, leveraging the experience gained during operation. For example, such methods include online convex optimization (OCO) [15], reinforcement learning (RL) [16] as well as bandit algorithms [17].

In the context of ISAC systems, the online learning methods proposed in the literature [18]–[24] are typically based on model-free RL that we later refer to as Model-Free Online Learning (MFOL) methods. The existing MFOL methods are typically sample inefficient when solving problems where the cardinality of the decision space is continuous or large, as typically is the case in ISAC systems. To address this problem, we proposed in [25] a MFOL method for resource allocation problems in multi-carrier ISAC systems where the decision space is continuous and combinatorial. The assumption about the action-independent state transition distribution improves its sample efficiency considerably. Thus it can be employed in real-time to solve resource allocation problems modeled as a constrained Markov decision processes (CMDPs) [26]. We also proposed a Model-Based Online Learning (MBOL) framework [27], [28] for resource allocation in ISAC systems. The MBOL methods utilize rich structural information on communications, sensing systems, and spectrum awareness. In addition, the MBOL approach aims to make learning more sample-efficient and explainable compared to the MFOL approach.

This paper provides an underlying theory for the MBOL framework and connects it to the MBOL algorithm proposed in [28]. We propose a specific class of MBOL methods and prove performance bounds describing their convergence properties. The analysis shows that the MBOL methods converge to the globally optimal policy under the assumption that the true environment dynamics can be captured perfectly with the used parametric model class. Otherwise, there is an asymptotic gap between the optimal and learned policies. The asymptotic gap is a function of the Kullback–Leibler (KL) divergence between the true model and the best model belonging to the model class. We also establish specific convergence rates for the proposed MBOL algorithm. Finally, the MBOL and MFOL methods are empirically compared in various interference scenarios, and the derived bounds are verified empirically. The results show that the empirical performance is considerably better than the derived performance bounds, i.e., the bound is not tight in the considered scenarios. This result aligns with related works in model-based RL, e.g., in [29]. Furthermore, the MBOL approach typically learns faster than the MFOL algorithm, and asymptotically the performance is not significantly worse compared to the MFOL. Thus, the explainability, data efficiency, and capability to exploit useful information make the MBOL approach attractive for solving resource allocation problems in ISAC applications.

This paper is organized as follows. Section II covers the related work in the context of online learning for ISAC systems. Section III introduces preliminaries for CMDPs that will be the basis for the rest of the paper. Section IV describes the specific class of MBOL methods considered in this paper. Section V introduces the ISAC waveform optimization problem and the MBOL algorithm for solving this problem. The performance bounds for the considered class of MBOL algorithms are analyzed in Section VI. Lastly, the performance of the proposed algorithms is evaluated numerically in Section VII.

II. Related Literature

Online learning methods, particularly multi-armed bandit (MAB) algorithms, have been extensively researched in the context of cognitive radios and opportunistic spectrum access; see [30] and references therein. Also, deep reinforcement learning (RL) has recently gained significant attention in wireless communications and networking context [31]. For example, a deep RL method for opportunistic spectrum access was proposed in [32]. In the context of radar and communications systems coexistence, an RL method to facilitate opportunistic spectrum access for cognitive radios operating in radar bands was proposed in [33]. In contrast, RL algorithms for cognitive radars operating in non-cooperative settings with communications have been proposed in [18]–[20]. In [18], a novel RL approach for non-cooperative coexistence was proposed, and [19] further extended the approach using deep RL algorithms. These RL algorithms select waveforms with different center frequencies and bandwidths from a library of waveforms. A contextual MAB algorithm for the non-cooperative setting was proposed in [20] to obtain better sample efficiency than the deep RL methods in [19]. For the cooperative setting, an RL algorithm was proposed in [21] to control the proportion of bandwidth shared between radar and communication systems.

In the context of co-designed integrated sensing and communications (ISAC) systems, a centralized deep RL method was proposed in [22] to allocate the power, frequency, and beam resources of multiple dual-function radar communication (DFRC) systems. Authors in [22] discretized the decision space and used a value-based deep RL algorithm to solve the problem. The optimization objective was based on the communications rate and mutual information (MI) for target detection. Also, in automotive applications, deep RL methods for scheduling radar and communications tasks in DFRC systems were proposed in [23], [24]. These methods learn to schedule the transceiver for sensing or communications purposes based on the road conditions and urgency of the transmitted communications data. In [24], a multi-agent learning setting is also considered.

III. Constrained Markov decision processes

A Markov decision process (MDP) [16] is composed of an agent making decisions sequentially in an environment. The decisions are called actions, and the possible actions the agent can take are specified by the action space $\mathcal{A}$. The environment is described by the state space $\mathcal{S}$ and the state transition distribution $\mathrm{T}(\mathbf{s}_{k+1}|\mathbf{s}_k, \mathbf{a}_k) \coloneqq \Pr(\mathbf{s}_{k+1}|\mathbf{s}_k, \mathbf{a}_k)$ where $\mathbf{s}_{k+1}, \mathbf{s}_k \in \mathcal{S}$, $\mathbf{a}_k \in \mathcal{A}$, and the subscript k denotes the time instance. The action and state spaces can be either discrete or continuous. The policy $\pi(\mathbf{a}|\mathbf{s}_k)$ is a probability distribution from which the agent samples actions given a specific state $\mathbf{s}_k$. For deterministic policies, we overload the notation such that $\mathbf{a}_k = \pi(\mathbf{s}_k)$ where the action is a deterministic mapping from a state. An important property of any MDP is that the optimal policy is deterministic [16].

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We concentrate on the *non-episodic* MDPs. Those start from an initial state distribution $\mu_0(\mathbf{s})$, but no terminal states exist. Assume that the Markov chain under policy π is ergodic such that we can define the stationary distribution as follows

$$\mu_\pi(\mathbf{s}) = \lim_{k\to\infty} p_\pi(S_k = \mathbf{s}|S_0 = \mathbf{s}_0)\ \forall\ \mathbf{s}_0. \tag{1}$$

The stationary distribution is independent of the initial state $\mathbf{s}_0$ but dependent on the policy π. Note that ergodicity is sufficient but not necessary for ensuring that the stationary distribution exists. However, it is common to assume the existence of the ergodic stationary distribution in RL literature [16, Chapter 10.3].

Consider a constrained Markov decision process (CMDP) with N_C constraints. The CMDP objective incorporates scalar random variables $u_{k,n} \sim \Pr(u_{k,n}\,|\,\mathbf{s}_{k-1}, \mathbf{s}_k, \mathbf{a}_{k-1})$ for all $n = 0, \ldots, N_C$ that are referred to as utilities. The utility $u_{k,0}$ corresponds to the rewards to maximize, and the utilities $u_{k,n}$ for all $n = 1, ..., N_C$ are used to formulate the constraints. For convenience, we make use of the following notation

$$u_n(\mathbf{s}_k, \mathbf{s}_{k+1}, \mathbf{a}_k) \coloneqq \mathbb{E}\left[u_{k+1,n}|\mathbf{s}_k, \mathbf{s}_{k+1}, \mathbf{a}_k\right] \tag{2}$$

as well as the marginal expectations of it, e.g., $u_n(\mathbf{s}_k, \mathbf{a}_k)$. The value function is the expected discounted sum of utilities starting from a state $\mathbf{s}$ that is defined as follows

$$V^\pi_{\gamma,n}(\mathbf{s}) = \mathbb{E}_\pi\left[\sum_{j=0}^{\infty} \gamma^j u_{k+j+1,n}\ |\ S_k = \mathbf{s}\right]. \tag{3}$$

where γ is the discount factor. In this paper, we study CMDP objectives that take the expected value of (3) with respect to the stationary distribution. Under the assumptions in (1) (see [16] for the proof)

$$\mathbb{E}_{\mathbf{s}\sim\mu_\pi}\left[V^\pi_{\gamma,n}(\mathbf{s})\right] = \frac{1}{1-\gamma} J_n(\pi), \tag{4}$$

where

$$J_n(\pi) \coloneqq \mathbb{E}_{\mu_\pi}\left[u_n(\mathbf{s}_k, \mathbf{s}_{k+1}, \mathbf{a}_k)|\pi\right]. \tag{5}$$

Thus, we may write our CMDP objective as follows

$$\arg\max_{\pi}\ J_0(\pi) \tag{6a}$$

$$\text{s.t. } J_n(\pi) \geq C_n\ \forall\ n = 1, \ldots, N_C \tag{6b}$$

where C_n is the lower bound imposed by the constraint n.

IV. Model-based online learning (MBOL) methods

The reinforcement learning (RL) framework [16] includes a variety of methods for solving MDPs with unknown transition or reward models. Those methods can be coarsely divided into model-free [16] and model-based methods [34]. Model-free methods directly learn the policy π or the value function in (3) by trial and error from its experiences. Model-based methods, on the other hand, learn the transition model $\mathrm{T}(\mathbf{s}_{k+1}|\mathbf{s}_k, \mathbf{a}_k)$ using its experiences and plan over the learned model as illustrated in Fig. 1. The planning in model-based methods is typically realized by two approaches: (i) methods that learn a global policy (similar to model-free algorithms) by augmenting real-world data with data generated by the model, and (ii) model predictive control (MPC) methods that optimize actions online. The latter is called the learning-based model predictive control (LMPC) approach [35]. This approach has several desirable properties in the context of ISAC applications. For example, the side information from the cooperative systems can be directly incorporated into the model. Furthermore, learning is not necessarily required when constraints or reward functions change.

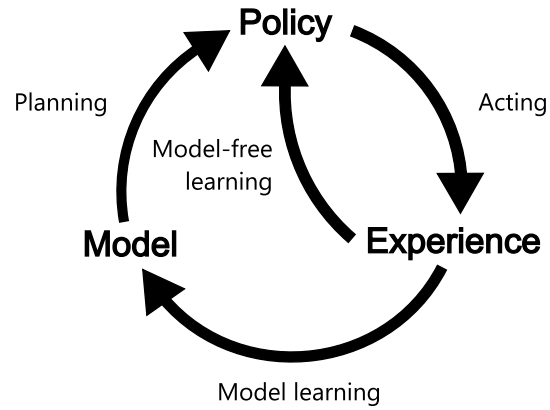

Fig. 1: Difference on model-free and model-based online (reinforcement) learning methods [16].

A. Proposed class of MBOL methods

The Model-Based Online Learning (MBOL) methods proposed in this paper stem from the LMPC approach. However, we consider a subset of CMDPs problems where the following assumptions hold.

Assumption 1. *The transition distribution is independent of the actions such that* $\mathrm{T}(\mathbf{s}_{k+1}|\mathbf{s}_k, \mathbf{a}_k) = \mathrm{T}(\mathbf{s}_{k+1}|\mathbf{s}_k)$ *for all* $\mathbf{s}_k \in \mathcal{S}$, $\mathbf{s}_{k+1} \in \mathcal{S}$ *and* $\mathbf{a}_k \in \mathcal{A}$.

Assumption 2. *Utility functions* $u_n(\mathbf{s}_k, \mathbf{s}_{k+1}, \mathbf{a}_k)$ *of the considered CMDP problem are known for all* $n = 0, \ldots, N_C$.

In ISAC applications, Assumption 1 is justified in contextual scenarios where interference and channel coefficients are independent of the resource allocations of our ISAC agent. However, the proposed MBOL method can be used in a reactive environment, where Assumption 1 is violated, as shown in [28], but there are no performance guarantees. A better way to address reactive interference would be to employ the full RL framework [16], especially when intelligent adversary agents and smart jammers are encountered. This could be a topic of study in future research, especially in the context of security and defense applications. However, developing similar theoretical insights and guarantees for the MBOL performance as this paper proposes is challenging in the more general setting. Lastly, Assumption 2 is justified since utilities can be typically expressed as functions of interference and channel states, for example, through signal-to-interference-plus-noise ratio (SINR).

The CMDP objective uses the average utility functions defined in (5). However, under Assumption 1 we can write the

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joint stationary distribution as follows

$$\mu(\mathbf{s}_k, \mathbf{s}_{k+1}) := \mathrm{T}(\mathbf{s}_{k+1}|\mathbf{s}_k)\mu(\mathbf{s}_k). \tag{7}$$

that is independent of the policy π. The LMPC-based approaches do not directly optimize (6). Let us define the expected utility under the model T_θ as follows

$$u_n^\theta(\mathbf{s}, \mathbf{a}) := \mathbb{E}_{\mathbf{s}_{k+1}\sim \mathrm{T}_\theta}\left[u_0(\mathbf{s}_k, \mathbf{s}_{k+1}, \mathbf{a}_k)|\mathbf{a}_k = \mathbf{a}, \mathbf{s}_k = \mathbf{s}\right] \tag{8}$$

where T_θ is the transition model with parameters $\theta \in \Theta$ to be learned. The LMPC policy $\pi_\theta(\mathbf{s}_k)$ optimizes a surrogate problem

$$\underset{\mathbf{a}\in\mathcal{A}}{\arg\max}\ u_0^\theta(\mathbf{s}_k, \mathbf{a}) \tag{9a}$$

$$\text{s.t. } u_n^\theta(\mathbf{s}_k, \mathbf{a}) \geq C_n\ \forall\ n = 1, \ldots, N_C. \tag{9b}$$

In the constrained setting, the objective in (9) is a suboptimal surrogate for (6) even if $T = T_\theta$. This is the case because the constraint in (9b) is more strict than the constraint in (6b) due to the conditioning on the current state. However, optimizing (9) also satisfies the constraint in (6) under the assumption that the constraint in (9b) can be satisfied for all $\mathbf{s}_k \in \mathcal{S}$.

This paper considers the model learning in LMPC by minimizing the Kullback–Leibler (KL) divergence between the true transition distribution T and the model T_θ. The KL divergence conditional to the current state is defined as follows

$$\mathcal{D}_{\mathrm{KL}}(\theta, \mathbf{s}) := \mathbb{E}_{\mathbf{s}'\sim\mathrm{T}}\left[\log\frac{\mathrm{T}(\mathbf{s}'|\mathbf{s})}{\mathrm{T}_\theta(\mathbf{s}'|\mathbf{s})}\middle|\mathbf{s}\right]. \tag{10}$$

Thus the loss function for the model learning can be written as follows

$$L(\theta) = \mathbb{E}_\mu\left[l(\theta; \mathbf{s}, \mathbf{s}')\right], \tag{11}$$

where

$$l(\theta; \mathbf{s}, \mathbf{s}') = \xi - \log \mathrm{T}_\theta(\mathbf{s}'|\mathbf{s}). \tag{12}$$

In (12), $\xi = \mathbb{E}_\mu\left[\log \mathrm{T}(\mathbf{s}'|\mathbf{s})\right]$ is a constant and does not affect the learning. By observing a sequence of states, an online learning algorithm $\mathcal{U}$ (e.g., an online convex optimization (OCO) [15] algorithm) is employed as follows

$$\theta_{k+1} = \mathcal{U}(\theta_k, \mathbf{s}_k, \mathbf{s}_{k+1}) \tag{13}$$

to update the model parameters. The parameters are updated such that

$$\bar{\theta}_K = \frac{1}{K}\sum_{k=1}^{K}\theta_k \tag{14}$$

minimizes the loss $L(\bar{\theta}_K)$ incrementally. In practice, θ_K would also minimize $L(\theta_K)$, but we are able to analyze the convergence only for (14). In Section VI-A, the convergence bounds for this algorithm are established.

V. MBOL for ISAC waveform optimization

This section presents the MBOL framework in ISAC resource allocation and waveform optimization problems. Consider a colocated and co-designed multi-carrier ISAC system operating in a radio spectrum with varying interference and channel conditions. The scenario captures a co-designed ISAC system, a communications user transmitting and receiving data from the ISAC system, and cooperative, non-cooperative and adversarial communications or sensing systems shown in Fig. 2. The cooperative system represents a system that can share information about the channel state, noise, and interference statistics with the ISAC system. In contrast, no information is shared or exchanged with the non-cooperative system. Lastly, the adversarial system represents intentional interference sources. Our goal is to continuously improve the performance of the multi-carrier ISAC system in dynamic scenarios by actively allocating the frequency and power resources. In other words, sub-carriers are selected and power allocated for sensing and communications to optimize the performance of the tasks at hand. The objective may be to maximize the sensing performance while satisfying certain constraints for the communications task or vice-versa. Alternatively, we could solve an unconstrained multi-objective problem.

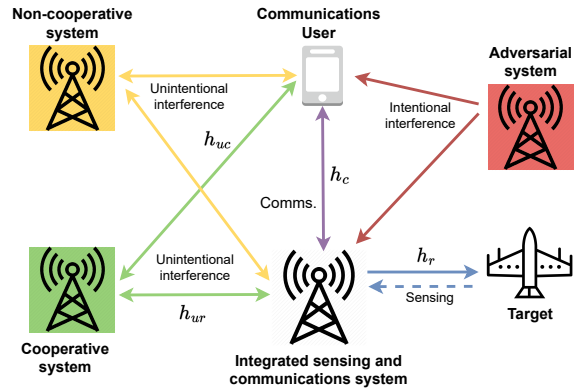

Fig. 2: Integrated sensing and communications (ISAC) system operating in coexistence with non-cooperative, cooperative and adversarial systems.

A. Signal model for waveform optimization

The interference and channel statistics are assumed quasi-stationary, i.e., wide-sense stationary (WSS) over the observation period. We refer to each observation period with time slot indices $k \in \mathbb{N}$ where the time t in slot k is in $[t_k, t_{k+1}]$. Consider a joint sensing and communications signal $y_r(t) + y_c(t)$ where $y_r(t)$ is the sensing waveform and $y_c(t)$ is the communications signal. Let us denote the sensing receiver (RX) at the ISAC system and the communications user RX with the subscripts r and c, respectively. The system model and the notation are illustrated in Fig. 2. Thus, the signal at a RX $x \in \{r, c\}$ on each time slot k is written as follows

$$r_x(t) = \sum_{j=1}^{L_x} h_{x,j}(t)\left[y_r(t-\tau_{x,j}) + y_c(t-\tau_{x,j})\right] + \sum_{j=1}^{L_{ux}} h_{ux,j}(t) y_u(t-\tau_{ux,j}) + v_x(t) \tag{15}$$

where L_x, $h_{x,j}(t)$, and $\tau_{x,j}$ denote the number of resolvable paths, channel coefficient, and the propagation delay for the path j, respectively. In addition, $v_x(t)$ represents the receiver

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noise and the interference from the non-cooperative and adversarial systems, i.e., it contains all the disturbances. The statistics of $v_x(t)$ is modeled as colored Gaussian noise. The number of resolvable paths, channel, and propagation delay for the cooperative signal are denoted as L_{ux}, $h_{ux,j}(t)$, and $\tau_{ux,j}$, respectively. Although the shared information such as channels and signal covariance matrices from the cooperative system are utilized, the actual launched signal $y_u(t)$ is considered to be an interfering signal modeled as colored Gaussian noise.

To express the discrete-time RX signal model used to formulate the resource allocation problem, we first compensate for the line-of-sight path delay and remove the Doppler shift. Thus, the signal at the RX x is written as follows

$$\mathbf{r}_x = \mathbf{H}_x(\mathbf{y}_r + \mathbf{y}_c) + \mathbf{H}_{ux}\mathbf{y}_u + \mathbf{v}_x. \tag{16}$$

The matrices $\mathbf{H}_x$ and $\mathbf{H}_{ux}$ are Toeplitz matrices with dimension $N \times N$ where N is the number of sub-carriers. This paper utilizes the properties of Toeplitz matrices to approximate the signal model to facilitate a convenient waveform optimization model.[1] We especially assume that Toeplitz matrices can be approximated as circulant matrices [36]. This assumption is satisfied with equality in the asymptotic region but is a good approximation even with reasonable small matrix dimensions as shown in [37]. Typically, ISAC systems are wide-band systems with a large number of sub-carriers N; thus, we assume that the approximation is sufficient. Therefore, and possibly because of the cyclic prefix, the channel matrices in the frequency domain can be written as diagonal matrices $\mathbf{D}_x$ and $\mathbf{D}_{ux}$, where the diagonal elements are discrete Fourier transforms of the first columns of $\mathbf{H}_x$ and $\mathbf{H}_{ux}$, respectively. Since $\mathbf{y}_u$ and $\mathbf{v}_x$ in (16) are WSS noise with Topeliz covariance matrix structure, also their covariance matrices are approximated as circulant matrices. Therefore, the interference plus noise covariance matrix in the frequency domain is written as follows

$$\mathbf{\Omega}_x = \mathbf{D}_{ux}\mathbf{R}_u\mathbf{D}_{ux}^{\mathrm{H}} + \mathbf{\Psi}_x, \tag{17}$$

where $\mathbf{R}_u$ and $\mathbf{\Psi}_x$ are the diagonal frequency domain covariance matrices of the cooperative signal and disturbances, respectively. Using the aforementioned approximations, the RX signal covariance matrix defined in (17) is a diagonal matrix.

B. CMDP formulation

This section formulates the resource allocation problem as a CMDP. Here we assume that the state is fully observable and leave the partially observed setting for future work.

State: The state $\mathbf{s}_k \in \mathbb{R}_+^{2N}$ at time slot k is composed of the channel qualities for sensing and communications functions. The notation $\mathbf{s}_{k,r} \in \mathbb{R}_+^N$ and $\mathbf{s}_{k,c} \in \mathbb{R}_+^N$ is used for the sensing state and the communications state, respectively. Similarly, channel qualities at sub-carrier i are denoted as $s_{k,c,i}$ and $s_{k,r,i}$. The channel quality for time slot k, function $x \in \{r, c\}$, and channel i is defined in a linear scale as follows

$$s_{k,x,i} = \frac{g_{k,x,i}}{\sigma_{k,x,i}^2} \tag{18}$$

that is the channel gain divided by the interference plus noise power.

Action: The action $\mathbf{a}_k$ is composed of the sub-carrier selection $\mathbf{w}_k \in \{r, c\}^N$ and power allocation $\mathbf{p}_k \in \mathbb{R}_+^N$ vectors. For the power, we have a total power constraint $\mathbf{1}^{\mathrm{T}}\mathbf{p}_k \leq P_{\mathrm{tot}}$ and a spectral mask constraint $\mathbf{p}_k \leq \mathbf{c}$ where the latter is imposed for managing the interference caused to the cooperative users. We use the notation $\mathcal{A}_p$ and $\mathcal{A}_w$ for the power allocation and the sub-carrier selection action spaces, respectively.

Utility function: The utility for function $x \in \{r, c\}$ is defined as follows

$$u_x(\mathbf{s}_{k+1}, \mathbf{a}_k) = \sum_{i=1}^{N} \mathbb{1}_{\{w_{k,i}=x\}} \log\left(1 + s_{k+1,x,i}\, p_{k,i}\right). \tag{19}$$

where $\mathbb{1}_{\{w_{k,i}=x\}}$ is an indicator function and refers to selecting sub-carrier i to be used for the function x. For sensing, (19) is the MI between the received signal and the target scattering matrix [8], and for communications, it is the data rate expressed by the Shannon-Hartley formula.

The choice of state plays a crucial role as it provides the information used by the MBOL algorithm to select its actions. As outlined in [28], the state can be divided into channel gains and interference. This separation facilitates using different update and prediction models, as these elements are typically independent. Such decomposition can more effectively capture the true dynamics of the MDP and, for example, enable conventional filtering methods to track channel gains. In a collaborative setting, learning the unknown part of the interference and treating interference from the cooperative systems as known through feedback and sharing situational awareness [28] is possible. Consequently, the MBOL algorithm can immediately adapt if cooperative systems change their resource allocation strategy or are experiencing more interference from the ISAC agent running the MBOL algorithm. Furthermore, aggregating multiple past spectrum states may reflect the true Markovian state more accurately since spectrum state transitions are not necessarily Markovian [28]. Also, previous actions may be aggregated to the state to address reactive interference environments where the spectrum transitions may additionally depend on the previous actions [28]. In this paper, we focus mainly on addressing dynamic interference scenarios and assume action-independent transitions (Assumption 1) to simplify the theoretical analysis. We adopt the simplified state presentation (18) for clarity, but our models and methods apply to more complicated state representations.

The MI criterion (19) for sensing is widely used for waveform optimization in radar systems [38], and especially in the multicarrier ISAC waveform optimization [7]–[10]. It can be used for target recognition tasks [38], as a surrogate for the probability of detection at a fixed false alarm rate [39], as well as for multi-target target tracking [40], for example. There is also a direct connection between the mean square error (MSE) and the MI, as established in [41]. Furthermore, it directly contributes to improved range resolution due to its tendency to distribute power to multiple carriers when channel conditions are favorable. This is the case because the Cramér–Rao bound

[1]The approximations are used for developing the waveform optimization method but are not necessary for processing the sensing and communications signals.

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(CRB) depends on the root mean square (RMS) bandwidth; thus, the method should spread the power broadly over the bandwidth. However, optimizing MI may induce undesired peaks in the range of sidelobes, which may be an issue for some sensing tasks. The sidelobes can be mitigated at the RX using a mismatched filter [42]. Additionally, methods to deal with the range sidelobes at the waveform design level have been proposed in the literature, such as introducing the sub-carrier power ratio constraint [11]. These constraints can be introduced for MBOL algorithm affecting the action space. In the case of the Model-Free Online Learning (MFOL) method, including constraints for action space is less trivial. For the sake of simplicity and to facilitate easier comparison between the MBOL and MFOL methods, these additional constraints are not considered in this paper.

C. Proposed MBOL algorithm

For convenience, we consider only the objective where the sensing performance is maximized under the constraint on the minimum average communications MI, denoted by $C_{\min}$. Modifying the proposed algorithm to maximize communications performance under sensing constraints is trivial. Thus in the optimization objective (6), we define $J_0(\pi) = \mathbb{E}_\mu\left[u_r(\mathbf{s}_{k+1}, \pi_\theta(\mathbf{s}_k))\right]$, $J_1(\pi) = \mathbb{E}_\mu\left[u_c(\mathbf{s}_{k+1}, \pi_\theta(\mathbf{s}_k))\right]$, $C_1 = C_{\min}$, and $N_C = 1$.

1) Transition model: We assume that the transition distribution is independent among the channels. Thus by using the product rule for independent random variables, we can write the transition probability distribution as follows

$$\mathrm{T}_\theta(\mathbf{s}_{k+1,x}|\mathbf{s}_k) = \prod_{i=1}^{N} \mathrm{T}_\theta(s_{k+1,x,i}|\mathbf{s}_k). \tag{20}$$

Furthermore, we approximate the channel-wise probability distributions $\mathrm{T}_\theta(s_{k+1,x,i}|\mathbf{s}_k)$ with step approximations such that the probability distribution is piece-wise constant and has at maximum L number of probability density values. Therefore, we define L approximation bins $\{\mathcal{C}_j\}_{j=1}^{L}$ where $\mathcal{C}_j \subset \mathcal{S}_{x,i}$ and $\mathcal{S}_{x,i}$ is the state-space for function x and channel i. The collection $\{\mathcal{C}_j\}_{j=1}^{L}$ should be disjoint and cover the whole state-space $\mathcal{S}_{x,i}$. To simplify the notation, assume that the collection $\{\mathcal{C}_j\}_{j=1}^{L}$ is the same for each x and i. Then the channel-wise transition model is written as follows

$$\mathrm{T}_\theta(s_{k+1,x,i}|\mathbf{s}_k) = \prod_{j=1}^{L}\left[\frac{\phi_{x,i,j}(\mathbf{s}_k,\theta)}{\mathcal{V}_j}\right]^{\mathbb{1}_{\{s_{k+1,x,i}\in\mathcal{C}_j\}}} \tag{21}$$

where $\mathcal{V}_j$ is the volume of the set $\mathcal{C}_j$, and $\phi_{x,i,j}(\mathbf{s}_k,\theta)$ is the function to be learnt for all x, i, and j. Intuitively, $\phi_{x,i,j}$ is the probability that the state of the function x on channel i will be in the approximation bin $\mathcal{C}_j$. The probabilities are defined using a softmax function as follows

$$\phi_{x,i,j}(\mathbf{s}_k,\theta) = \frac{\exp f_{x,i,j}(\mathbf{s}_k,\theta)}{\sum_{l=1}^{N}\exp f_{x,i,l}(\mathbf{s}_k,\theta)} \tag{22}$$

where $\mathbf{f}_x : \mathcal{S}\times\theta \to \mathbb{R}^{N\times L}$ can be, for example, a feedforward neural network. Using the model defined above, the loss in (11) can be written as follows

$$L(\theta) = \xi - \sum_{x\in\{r,c\}}\sum_{i=1}^{N}\mathbb{E}\left[\sum_{j=1}^{L}\mathbb{1}_{\{s_{k+1,x,i}\in\mathcal{C}_j\}}\log\phi_{x,i,j}(\mathbf{s}_k,\theta)\right]. \tag{23}$$

This loss function shows that the model learning problem is identical to solving multiple multiclass logistic regression problems [43] for each sensing and communication function x and channel i.

2) Controller: The controller optimizes the surrogate defined in (9). For that, the expectations are computed as follows

$$\mathbb{E}_{\mathrm{T}_\theta}\left[u_x(\mathbf{a}_k,\mathbf{s}_{k+1})|\mathbf{a}_k,\mathbf{s}_k\right] \approx \sum_{i=1}^{N}\sum_{j=1}^{L}\mathbb{1}_{\{w_{k,i}=x\}} \frac{\phi_{x,i,j}(\mathbf{s}_k,\theta)}{2}\left[\log(1+\check{b}_j p_{k,i}) + \log(1+\hat{b}_j p_{k,i})\right] \tag{24}$$

where $\check{b}_j$ and $\hat{b}_j$ are the boundary values of the approximation bin $\mathcal{C}_j$. The equation in (24) is a concave function of the powers $\mathbf{p}_k$ (included in action $\mathbf{a}_k$). This concavity is useful from the perspective of convex optimization algorithms. However, the problem in (9) along with the utility functions defined in (19) is a non-convex mixed-integer problem. We relax the optimization problem by first selecting all channels to be communication channels and optimizing with respect to the powers. Then we find the minimum number of sub-carriers that satisfy the communications requirement. After that, the excess power not needed for satisfying the communications constraint is identified and reallocated for sensing by minimizing the power allocated to communications subject to the communications MI constraint. The remaining power budget is optimized for sensing function using the remaining sub-carriers. The pseudo-code for the controller algorithm is presented in Algorithm 1.

Algorithm 1: The controller algorithm

Input: Transition model T_θ, $\mathbf{s}_k$

```
// Optimize for communications
```

Set $w_{k,i} = c \;\forall\, i \in \{1,\ldots,N\}$;

Solve for $\mathbf{p}_{k,c} \in \mathcal{A}_p$ by maximizing $\mathbb{E}_{\mathrm{T}_\theta}\left[u_c(\mathbf{a},\mathbf{s}_{k+1})|\mathbf{s}_k,\mathbf{w}_k\right]$;

Find minimum # of channels that satisfy (9b), set $\mathbf{w}_{k,i} = r$ for the remaining channels i;

Remove excess power from $\mathbf{p}_{k,c}$ such that (9b) is met exactly;

```
// Optimize for sensing
```

Compute the remaining power budget, set to P_r;

Solve for $\mathbf{p}_{k,r} \in \mathcal{A}_p$ by maximizing $\mathbb{E}_{\mathrm{T}_\theta}\left[u_r(\mathbf{a},\mathbf{s}_{k+1})|\mathbf{s}_k,\mathbf{w}_k\right]$ s.t. power budget P_r ;

Set $\mathbf{p}_k = \mathbf{p}_{k,c} + \mathbf{p}_{k,r}$;

return $\mathbf{p}_k$, $\mathbf{w}_k$

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VI. MBOL PERFORMANCE BOUNDS

Model-based RL algorithms have been analyzed in [29], [44], [45] (and references therein). In [29], a gap between the model-based and true value functions was obtained, and a monotonic improvement property of the model-based policy optimization algorithm was proven. However, performance bounds or the CMDPs were not considered. In [44], [45], regret bounds for value-targeted model-based RL were provided under the assumption of linear or mixture models. In this paper, we consider transition models that can be non-linear for which the analysis in [44], [45] is not applicable. We analyze the class of MBOL problems proposed in Section IV-A.

A. General MBOL performance bounds

This section establishes three performance bounds for model convergence, policy convergence, and constraint satisfaction.

1) Model convergence: The model convergence bound will be an integral part of the rest of the analysis done in this paper. Stochastic performance bounds of online learning algorithms can be obtained using the online-to-batch conversion when the data is independent and identically distributed (i.i.d.) [15]. However, the conversion does not apply to the OCO algorithm $\mathcal{U}$ in (13) because we have dependent data. Fortunately, the model convergence bounds can be analyzed using the theorem proposed in [46] where the generalization ability of OCO algorithms is analyzed for dependent data.

Before we can state the theorem in [46], we need to define the mixing coefficient used to measure the Markov chain convergence and describe the required assumptions.

Definition 1. *The β-mixing coefficient [46] of the sampling distribution T is defined as follows*

$$\beta(k) \coloneqq \sup_{t\in\mathbb{N}} \left\{2\mathbb{E}\left[\mathcal{D}_{\mathrm{TV}}\left(\mathrm{T}^{t+k}(\cdot|\mathcal{F}_t), \mu\right)\right]\right\} \tag{25}$$

where μ is the stationary distribution of T, T^{t+k} denotes the distribution of $\mathbf{s}_{t+k}$, and $\mathcal{F}_t$ is sigma-field of observed states $\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_t$.

We say that the distribution T is β-mixing if $\lim_{k\to\infty}\beta(k) = 0$. Next, the following assumptions need to be satisfied.

Assumption 3. *Loss function $l(\theta; \mathbf{s}, \mathbf{s}')$ in (12) is a convex function w.r.t. parameters $\theta \in \Theta$, and G-Lipschitz w.r.t. norm $\|\cdot\|$ over Θ*

$$|l(\theta; \mathbf{s}, \mathbf{s}') - l(\theta^*; \mathbf{s}, \mathbf{s}')| \leq G\|\theta - \theta^*\|. \tag{26}$$

Assumption 4. *The parameter space Θ is compact and has a finite radius*

$$\|\theta - \theta^*\| \leq D \tag{27}$$

for any $\theta, \theta^* \in \Theta$,

Assumption 5. *(Stability) There is a non-increasing sequence α_k such that if θ_k and θ_{k+1} are successive iterates of the online algorithm $\mathcal{U}$ in (13), then $\|\theta_k - \theta_{k+1}\| \leq \alpha_k$.*

Consider a stable OCO algorithm $\mathcal{U}$ whose regret is bounded above by

$$\sum_{k=1}^{K} l(\theta_k; \mathbf{s}_k, \mathbf{s}_{k+1}) - l(\theta^*; \mathbf{s}_k, \mathbf{s}_{k+1}) \leq \mathcal{R}_K^{\mathcal{M}} \tag{28}$$

for any $\theta^* \in \Theta$. Finally, we can state the theorem proposed in [46] that shows the convergence in expectation for stable OCO algorithms.

Theorem 1. *(Model convergence) [46] Under Assumptions 3, 4 and 5, for any $\theta^* \in \Theta$ we have the model learning loss bounded in expectation from above by*

$$\mathbb{E}\left[L(\bar{\theta}_K)\right] - L(\theta^*) \leq \mathcal{E}(K; \mathcal{U}) \tag{29}$$

where $\bar{\theta}_K$ is defined in (14) and $\mathcal{E}(K;\mathcal{U})$ is defined as follows

$$\mathcal{E}(K;\mathcal{U}) = \frac{1}{K}\mathbb{E}\left[\mathcal{R}_K^{\mathcal{M}}\right] + \inf_{\tau\in\mathbb{N}}\left[\beta(\tau-1)GD + \frac{(\tau-1)G}{K}\left(2D + \sum_{k=1}^{K}\alpha_k\right)\right]. \tag{30}$$

Proof. Refer to [46] for the proof. □

Note that (30) differs from the bound in [46] by replacing $\beta(\tau)$ with $\beta(\tau-1)$. This is because, in [46], the beta mixing coefficient is defined for a random variable that would be in our case $\mathbf{x}_t = (\mathbf{s}_t, \mathbf{s}_{t+1})$. Thus the β-mixing coefficient of $\mathbf{x}_t$ is the β-mixing coefficient of $\mathbf{s}_t$ delayed by one. Note that the bound in Theorem 1 converges to zero asymptotically for any stable OCO algorithm with sub-linear regret under β-mixing environments. An interested reader may also refer to [46], where high-probability convergence guarantees are derived. However, those are not considered in this paper, but the results directly apply to the problem considered in this paper.

2) Policy convergence in unconstrained setting: Define a shorthand notation $\pi_k \coloneqq \pi_{\bar{\theta}_k}$ where $\pi_{\bar{\theta}_k}$ is the MBOL policy at time slot k. We make the following assumption about the utility function that is used in analyzing the policy convergence properties for unconstrained settings.

Assumption 6. *The utility function is bounded $u_n(\mathbf{s}_k, \mathbf{s}_{k+1}, \mathbf{a}_k) \in [0, u_{\max}]$ for all $\mathbf{s}_k, \mathbf{s}_{k+1} \in \mathcal{S}$ and $\mathbf{a}_k \in \mathcal{A}$.*

Then we state the following theorem to prove the convergence properties of the MBOL algorithms in the unconstrained setting.

Theorem 2. *(Policy converge in unconstrained setting) Under Assumptions 3 to 6, the difference between the average rewards for the optimal policy π^* and the MBOL policy π_K is bounded in expectation from above by*

$$J_0(\pi^*) - \mathbb{E}\left[J_0(\pi_K)\right] \leq 2\sqrt{2}u_{\max}\sqrt{\Delta + \mathcal{E}(K;\mathcal{U})} \tag{31}$$

where $\Delta = \min_\theta L(\theta)$ is the minimum average KL divergence achievable by the model T_θ over Θ.

Proof. Given in the Appendix A. □

Since $\lim_{K\to\infty}\mathcal{E}(K;\mathcal{U}) = 0$, Theorem 2 states that MBOL algorithms converge to the optimal policy only if there exists $\theta^* \in \Theta$ such that $T = T_{\theta^*}$ inducing $\Delta = 0$. In addition, Theorem 2 shows that the convergence rate is in the worst-case worse than the convergence rate of the model because of the square root in the right-hand side of (31).

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3) Constraint satisfaction: Lastly, we will analyze the constraint satisfaction of the MBOL algorithms. For that, we make the following assumption about the controller in (9).

Assumption 7. *The model-based controller optimizing (9) always satisfies the constraints in (9b). In other words, there exists $\mathbf{a}^* \in \mathcal{A}$ such that $u_n^{\theta}(\mathbf{s}, \mathbf{a}^*) \geq C_n$ for all $\mathbf{s} \in \mathcal{S}$ and $n = 1, \ldots, N_C$ where C_n is the lower bound constraint defined in (6b).*

Now we can state the last theorem quantifying the MBOL algorithm performance.

Theorem 3. *(Constraint satisfaction) Under Assumptions from 3 to 7, MBOL algorithms obtain average constraint values that are bounded in expectation from below by*

$$\mathbb{E}\left[J_n(\pi_K)\right] \geq C_n - u_{\max}\sqrt{2(\Delta + \mathcal{E}(K;\mathcal{U}))}. \tag{32}$$

Proof. Given in the Appendix A. □

Theorem 3 shows that the MBOL algorithm does guarantee the constraints to be satisfied asymptotically only when $\Delta = 0$. In addition, the rate that this lower-bound would converge to the requirement C_n depends on the model-convergence rate similar to Theorem 2.

B. Problem dependent performance bounds

In this section, we analyze the exact model learning convergence rates of Theorem 1 for the problem proposed in Section V. Consequently, this analysis results in specific convergence rates for the bounds in Theorems 2 and 3. To do that, we also need to consider a specific OCO algorithm used for the model learning problem. We consider the Online Gradient Descent (OGD) algorithm, and its pseudo-code is shown in Algorithm 2. For the model proposed in Section V-C1, we

Algorithm 2: Online gradient descent [47]

Input: Parameter space Θ, initial parameters $\theta_1 \in \Theta$ and initial state $\mathbf{s}_1$

for k=1,...,K **do**

  Let $\nabla_k := \frac{\partial}{\partial\theta} l(\theta_k; \mathbf{s}_k, \mathbf{s}_{k+1})$;

  $\tilde{\theta}_{k+1} = \theta_k - \frac{D}{G\sqrt{k}}\nabla_k$;

  $\theta_{k+1} = \min_{\theta\in\Theta} \left\|\theta - \tilde{\theta}_{k+1}\right\|_2$

redefine the probabilities in (22) as follows

$$\phi_{x,i,j}(\mathbf{s}, \theta) = \frac{\exp \theta_{x,i,j}^{\mathrm{T}} \boldsymbol{\psi}(\mathbf{s})}{\sum_{l=1}^{L} \exp \theta_{x,i,l}^{\mathrm{T}} \boldsymbol{\psi}(\mathbf{s})}. \tag{33}$$

In (33) $\boldsymbol{\psi} : \mathcal{S} \to \mathbb{R}^M$ is a known mapping that may be non-linear. This model and the loss function (23) correspond to the linear multiclass logistic regression problem. It is well known that it is a convex problem. In addition, we will assume that the state space is bounded which induces Assumption 6. Also, the bounded state-space and the convexity induce Assumption 3.

Theorem 1 require the regret bound $\mathcal{R}_K^{\mathcal{M}}$, $\alpha_k \; \forall \; k$, and the β-mixing coefficients to be known for analyzing the exact convergence rates. Under Assumption 4, regret of the OGD algorithm is bounded by $\mathcal{R}_K^{\mathcal{M}} = \frac{3}{2}GD\sqrt{K}$ [15]. In addition, it is easy to show that $\alpha_k = Dk^{-\frac{1}{2}}$ that induces Assumption 5. Deriving the bound for β-mixing coefficient requires assuming certain mixing environments. To give an example of achievable converge rates, we assume that the true transition distribution belongs to the class of models proposed in (21). We prove in Appendix B that the model in (21) behaves similarly to finite state Markov chains. Furthermore, authors in [48] prove that convergence of ergodic finite state Markov chains can be bounded from above by $\beta(\tau) = \lambda\zeta^{-\tau}$ with some constants $\lambda > 0$ and $\zeta > 1$. Finally, we obtain a more specific bound for Theorem 1 such that

$$\begin{aligned}&\mathbb{E}\left[L(\bar{\theta}_K)\right] - L(\theta^*) \\ &\quad \leq \frac{GD}{K}\left[\frac{3}{2}\sqrt{K} + \lambda + 2\lceil\log_\zeta K\rceil\left(1 + \sqrt{K}\right)\right]\end{aligned} \tag{34}$$

where $\lceil\cdot\rceil$ rounds up to the nearest integer. The bound in (34) is obtained from (30) by setting $\tau = \lceil\log_\zeta K\rceil + 1$.

The bound in (34) shows that the convergence rate for the model learning problem is of the order $O(\frac{GD\log_\zeta K}{\sqrt{K}})$. Furthermore, the policy and constraint convergence depend on this bound via the square-root term in the expressions of Theorems 2 and 3. Thus, the policy and constraint satisfaction convergence rates are of the order $O\left(\frac{u_{\max}\sqrt{GD\log_\zeta K}}{K^{\frac{1}{4}}}\right)$ to the locally optimal policy if $\Delta > 0$ or globally optimal policy if $\Delta = 0$. However, if the constant ζ is large, we approach the i.i.d. bound.

VII. Numerical Examples

We empirically evaluate the average sensing and communications MI performance of the MBOL algorithm proposed in Section V-C. We compare the proposed method to the previously proposed MFOL method [25], an efficient model-free algorithm tailored for solving the particular problem considered in this paper. It directly learns the resource allocation policy π by only observing the sequence of states. In addition, we compare the performance against three other baseline algorithms. To make the comparison fair, all baseline methods satisfy the total power P_{tot} and spectral mask $\mathbf{p}_k \leq \mathbf{c}$ constraints. A baseline referred to as "non-predictive" uses the current state observation as the best estimator for the upcoming state and optimizes the allocation accordingly at each time slot k. This algorithm corresponds to the structured optimization algorithm proposed in [8], where stationary channel and interference statistics are assumed. Another baseline based on structured optimization is referred to as the "known model" that has been given the parameters of the dynamics model. It uses similar optimization procedures introduced in Section V-C2 to allocate the resources actively. This baseline was developed to illustrate the achievable performance using the MBOL method with perfect model knowledge. Our third baseline, namely "uniform", uses a uniform power allocation among the sub-carriers except for the specific sub-carriers where constraint $\mathbf{p}_k \leq \mathbf{c}$ is strictly active. Furthermore, for each time step k and sub-carrier n, a sub-carrier is selected for sensing with probability 0.5; otherwise, it is allocated for communications.

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This baseline corresponds to ISAC systems that do not actively optimize the resource allocations for the underlying channel and interference conditions.

Two variants of the MBOL algorithms are considered. One uses the model in (33), and the other uses the model in (22), where the function $f_x : \mathcal{S}_x \times \theta_x \to \mathbb{R}^{N\times L}$ is a neural network. Based on the terminology used in multiclass logistic regression literature, those methods are later referred to as *linear MBOL* and *non-linear MBOL*, respectively. Both MBOL algorithms use $L = 8$ approximation bins defined in a logarithmic scale between 0.03 and 30. Throughout this section, the linear MBOL uses mapping $\boldsymbol{\psi} : \mathcal{S}_x \to \{0,1\}^{NL}$. The elements of the mapping $\boldsymbol{\psi}(\mathbf{s}_{k,x})$ are defined as follows

$$\psi_{(i-1)L+j}(\mathbf{s}_{k,x}) = \mathbb{1}_{\{s_{k,x,i}\in\mathcal{C}_j\}} \tag{35}$$

where $i \in \{1,\dots,N\}$ is the channel index and $j \in \{1,\dots,L\}$ is the approximation bin. Intuitively, this mapping one-hot encodes the state of each channel to the used approximation bins $\mathcal{C}_j$. To be consistent with our theoretical analysis in VI, for linear MBOL, we consider a parameter space with a bounded radius (Assumption 4). The parameter space considered is $\Theta = \{\theta \in \mathbb{R}^{N^2L^2} : |\theta_l| \leq \rho \ \forall\ l\}$ whose diameter is bounded above by $D = \rho NL$. Additionally, we use the OGD algorithm with the optimal learning rate that minimizes the upper bound of the model learning regret [15]. Therefore, the learning rate is $\frac{G}{D\sqrt{k}} = \frac{\rho\sqrt{L}}{k}$ where $G = N\sqrt{L}$ (see Appendix C). In all the considered numerical examples, we have $\rho = 4$.

The non-linear MBOL method uses feedforward neural networks (FNNs) with a single hidden layer of size $4N$ and the rectified linear unit (ReLU) activation function. The parameters are optimized using the *Adam* method [49], which is typically used to train neural networks. The MFOL algorithm uses a FNN with a single hidden layer of size $8N$, ReLU activations, and the *Adam* optimizer. In addition, the Lagrange multiplier of the MFOL method is updated using OGD with a constant learning rate of 1e-3. In the case of non-linear MBOL and MFOL methods, we cannot use the OCO framework for selecting the learning rates because the optimization problems are non-convex. However, the default parameters proposed in [49] worked well in our experiments. Note that the online learning setting differs from offline machine learning because one typically does not have the opportunity to experimentally tune the learning rate parameter for the incoming data because the data is not available beforehand.

A. Performance in constrained settings

Four different interference environments are considered in this section:

1) "Matched": The transition model is similar to that used by the linear MBOL algorithm, and the model parameters are sampled independently from the standard Gaussian distribution.
2) "Markov chain": A two-state Markov chain model is used for each channel to generate state transitions between idle and transmit states. When transmitting, we sample the state uniformly from the state space.
3) "Stationary": The state does not change, i.e., the interference and channel statistics are time-independent.
4) "Hopping": A frequency hopping pattern is generated via $\mathbf{s}_{x,k+1} = \mathrm{clip}(\mathbf{F}\mathbf{s}_{x,k} + \mathbf{v}_{x,k})$ where $\mathbf{F}$ is an off-diagonal matrix and $\mathbf{v}_{x,k}$ is zero-mean Gaussian noise with covariance $2\,\mathbf{I}$. The operator $\mathrm{clip}(\cdot)$ clips the input to range $[0.03, 30]$.

In all scenarios, the communications requirement is set to $C_{\min} = 5$, the number of sub-carriers is $N = 16$, and the channel qualities vary between 0.03 and 30. The total power P_{tot} of the ISAC system is 5. The cooperative system imposes the constraint vector $\mathbf{c}$ to restrict three sub-carriers from the edges to have power below 1% of the total power. All the learning and baseline methods are simulated over 40 independent runs. The average MIs are approximated at each evaluation stage with 512 independent samples of the tuple $(\mathbf{s}_k, \mathbf{s}_{k+1})$ from the joint stationary distribution μ.

Fig. 3a shows the sensing performance under the four different interference scenarios. Let us first look at the performance in the "matched" scenario. The sensing performance of the linear MBOL method is the best. On the other hand, the sensing performance of the non-linear MBOL and MFOL are nearly identical but distinctively below the linear MBOL. To fully understand the performance in a constrained setting, we must also look at the communications performance in Fig. 3b. The linear MBOL method does not satisfy the communications requirement through the learning period in the "Matched" scenario. However, the performance is sufficiently close to the requirement (the MI is above 4.5) already after 1000 training samples, and it converges toward the requirement as the number of training samples increase. The behavior of the MBOL methods converging to the communications requirement from below is shown in Theorem 3. In contrast, the non-linear MBOL method satisfies the requirement immediately, and for the MFOL method, it takes less than 50 samples to reach sufficiently good performance. However, at the end of learning, the linear MBOL method is the only method with nearly as good performance as the method using the known model.

Next, we will look at the performance in the "Markov chain" scenario. In that scenario, the linear MBOL algorithm behaves similarly as in the "Matched" scenario, as seen in Fig. 3. Meaning that the sensing performance of the method increases steadily, and the communications performance converges towards the requirement from below. However, the non-linear MBOL satisfies the communications requirement almost immediately while obtaining a similar sensing performance to the linear variant. The sensing performance of the MFOL method in Fig. 3a is poor for the first one thousand training samples. Fig. 3b shows that this is because it wastes its resources on the communications function. After the communications performance is close to the requirement, the sensing performance improves rapidly. In fact, the performance is the best at the end of learning. It even surpasses the performance of the baseline using the known model.

The state transitions in the "hopping" scenario have less variance. Therefore, the number of samples required to learn a sufficiently good policy is smaller, as seen in Fig. 3. In this scenario, the MBOL-based methods converge fast to good

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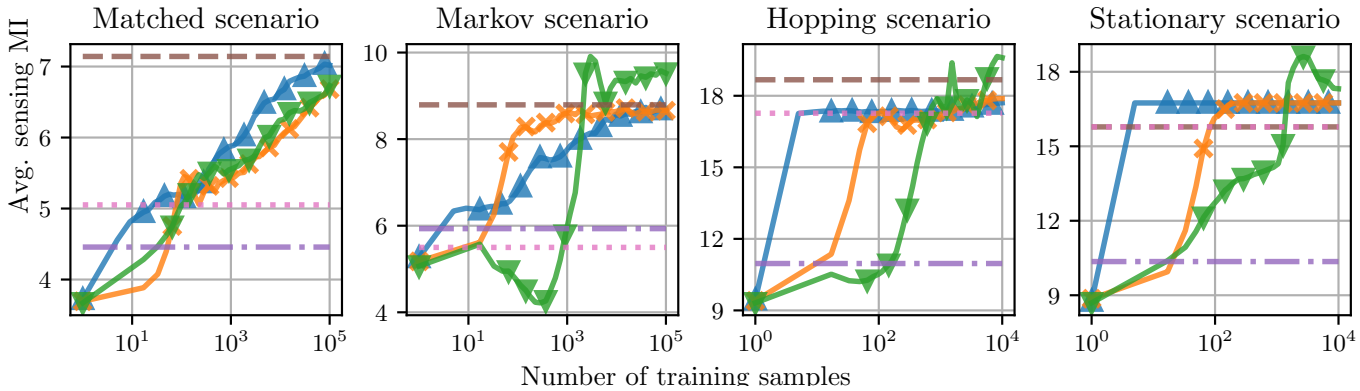
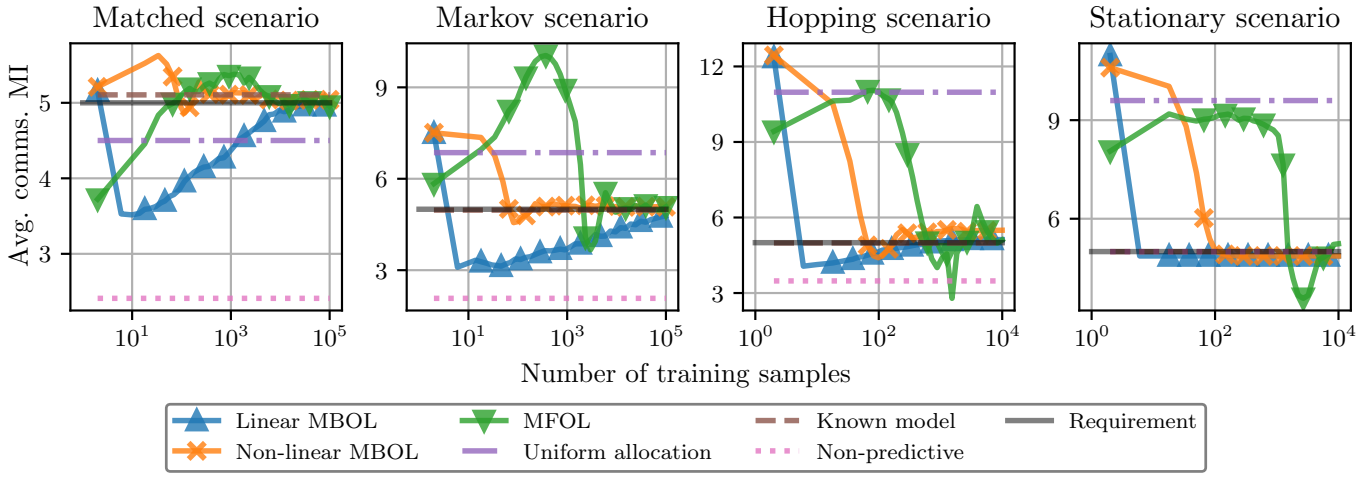


(a) Sensing performance throughout the learning period.



(b) Communications performance throughout the learning period.

Fig. 3: Comparing the proposed model-based online learning (MBOL) method to the model-free online learning (MFOL) method and the baseline algorithms.

performance, approximately after 100 samples. On the other hand, for the MFOL method, it takes a little over 1000 samples to converge similar performance. Moreover, we can observe that the MFOL suffers an oscillation in both sensing and communications performance. This behavior was also noticed in [25].

Lastly, in the "stationary" scenario, all the methods converge sufficiently fast. For example, the linear MBOL reaches its stable point after ten learning steps. However, the communications performance is slightly below the constraint, which indicates the loss due to the step approximation. We could mitigate this loss by increasing the number of approximation bins L, but it slows the learning speed, as shown analytically in Appendix C. The MFOL method achieves similar performance to MBOL methods approximately at 1000 learning steps. At the end of learning, the performance is even better than the "non-predictive" and "known model" baselines have (those methods are equal in the "stationary" scenario). Indirectly this shows that the MBOL controller proposed in Section V-C2 is not optimal. Furthermore, the performance of the MFOL method can surpass the performance of this controller even when the model is fully known due to the modeling error.

It can be seen that generally, the MFOL method performs better than the MBOL methods with a large number of samples in all except "Matched" scenarios. This can be explained based on the derived theoretical results. If the model bias (Δ in equation (31)) is not zero, the model cannot perfectly represent the true environment asymptotically. In such cases, the proposed algorithm will be suboptimal even in an asymptotic regime. Another reason the MBOL performance may be degraded is due to the sub-optimality of the controller. The MFOL method is not learning a model and instead directly tries to learn a policy that maximizes the performance objective. The simulations show this gives the flexibility to learn a good policy given the large amount of data. However, the best performance in the large sample region may not be useful because the radio environment is time-frequency-location varying. Also, the learning will be a black box and hard to explain or provide quantitative information about the performance.

B. Empirical convergence rate

In this section, we empirically evaluate the performance in an unconstrained setting and compare it to the baseline algorithms and the derived bounds in Section VI. This section considers the "Matched" scenario from Section VII-A with similar parameterization, except we set $C_{\min} = 0$.

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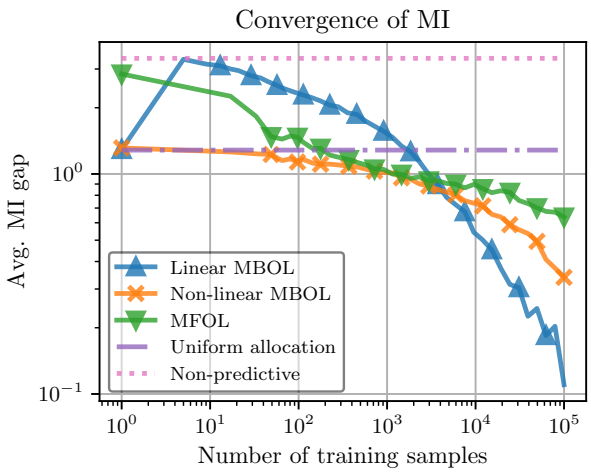
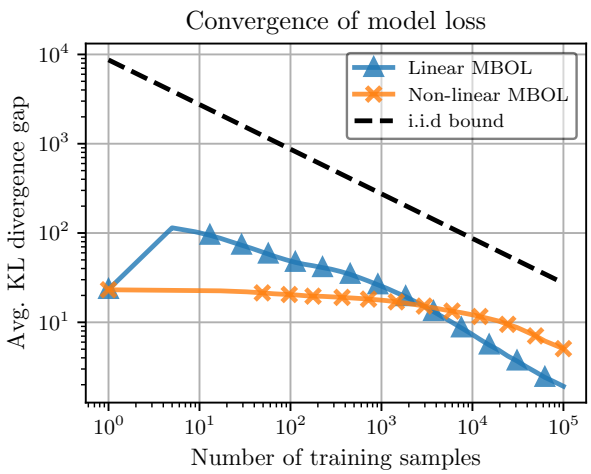


(a) Convergence of the MI optimization criterion.



(b) Convergence of the model loss function for the model-based algorithms.

Fig. 4: Empirical convergence analysis of the algorithms.

Fig. 4a shows the convergence of the average MI for the sensing function, i.e., the y-axis is the difference between the optimal MI and the obtained MI. Initially, the performance of the MBOL-based algorithms is equal to the performance of the "uniform allocation" baseline. The linear MBOL diverges from the initial guess for the first ten training samples but, after that, starts converging. The non-linear MBOL method starts immediately to converge, as seen in Fig. 4a. However, the convergence rate is slower. The performance of linear and non-linear methods overlap approximately at 3000 samples. The MFOL algorithm performs the worst compared to the other learning methods, as seen in Fig. 4a.

Surprisingly, the linear MBOL method initially performs worse than the other learning methods. A similar phenomenon can be seen in Fig. 4b, which shows the average KL divergence. This coupling between the MI and the model learning loss is expected by Theorem 2. However, the divergence at the beginning may be related to the optimal choice of the learning rate parameter. The learning rate is large enough that the OGD algorithm can effectively explore the whole parameter space Θ,

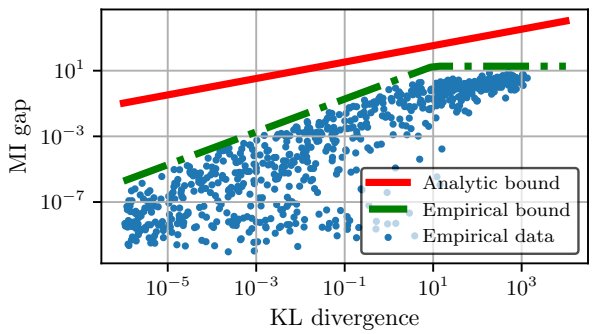


Fig. 5: Bound between optimal utilities and utilities obtained using the model-based approach with different KL-divergences between the real transition distribution and the model.

which may initially lead the parameter estimates far away from the optimal parameters. The learning rate of the non-linear method is effectively lower than the linear method, which prevents significant model updates at the beginning of learning. However, we observed that increasing the learning rate makes the non-linear algorithm diverge. We can not guarantee that the non-linear method can converge to the optimal policy because the model learning problem is non-convex. Fig 4a shows that the non-linear MBOL and the MFOL methods have slower convergence rates compared to the linear MBOL method.

We empirically observed that the bounds derived in Section (VI) are not tight in the considered scenario. There are two reasons for that. The first reason is the regret bound for the model loss $\mathcal{R}_K^{\mathcal{M}}$ introduced in Section VI-B. This bound is based on OCO framework and applies to stochastic loss and adversarial loss functions [15]. Since the model learning problem is stochastic, the empirical performance is considerably better than the worst-case bound. The second and more significant reason for the lack of tightness is Lemma 2 in Appendix A. It is a bound between the optimal average sensing MI and the obtained MI using the MBOL algorithm described in Section IV-A. Fig. 5 visualizes this bound empirically by randomly sampling the true environment model and another model used by the MBOL algorithm. This model used for decision-making has a specific KL divergence from the true model. Fig. 5 shows that the analytic bound is, at best, 30dB larger than the empirical data would suggest. Moreover, this gap is even more significant when looking at the smaller KL divergence values. The empirical bound shown in Fig. 5 is fitted to the data with a linear relationship between the MI gap and the KL divergence. This bound becomes constant because Assumption 6 states that the MI (i.e., the utility) is bounded from above by $u_{\max}$, which is also the largest possible gap. In fact, this same upper bound could be used in the analytic bound, but it does not affect the derived convergence rates. This linear relationship indicates that policy convergence rates of order $O(\frac{\log K}{\sqrt{K}})$ (instead of $O(\frac{\log K}{K^{1/4}})$) can be achieved in practice.

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C. The effect of dependent data in model learning

In the final experiment, we demonstrate the effects of dependent data on model learning. Without loss of generality, we consider only a single-channel model to simplify the simulation setup. A Markov chain with L states is used, where the transition probability from bin $\mathcal{C}_j$ to $\mathcal{C}_i$ for all $i, j \in \{1, ..., L\}$ is

$$\mathrm{T}(s_{k+1} \in \mathcal{C}_i | s_k \in \mathcal{C}_j) = \begin{cases} 1-\epsilon & \text{if } i = j \\ \frac{\epsilon}{L-1} & \text{otherwise.} \end{cases} \tag{36}$$

This model is contained by the proposed linear MBOL model in (33) when using the mapping ψ in (35). The stationary distribution of this Markov chain is uniform, and ϵ affects the mixing time. The mixing time τ_{mix} is defined here as the smallest k for which $\mathcal{D}_{\text{TV}}\left(T^k(\cdot|\mathbf{s}_0), \mu\right) \leq \frac{1}{4}$. For all experiments, we use the optimal learning rate with $\rho = 4$.

Fig. 6 shows the model converge for Markov chains with different mixing times. We can see that learning with dependent data will make learning slower in comparison to the i.i.d. case, especially at the beginning. Furthermore, as the mixing time decreases, the difference between dependent and i.i.d. cases is less evident. Because the loss functions are composed of two subsequent states, Theorem 1 suggests that there is a difference between the i.i.d. and $\tau_{\text{mix}} = 0$ cases. However, this experiment shows that the difference is negligible in practice.

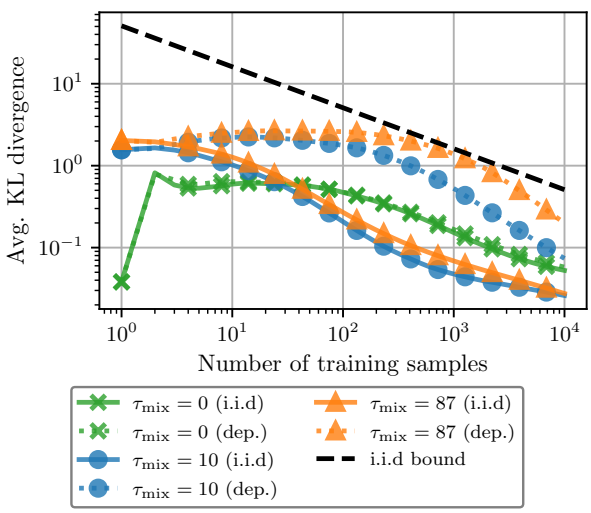

Fig. 6: Training models in environments with different mixing times. We can see that higher mixing time results in slower learning.

VIII. Conclusions

This paper introduced a class of Model-Based Online Learning (MBOL) methods that can be used for resource allocation and waveform optimization in integrated sensing and communications (ISAC) systems. We analyzed and proved convergence rates for policy convergence in unconstrained problems and constraint satisfaction in constrained problems. We demonstrated the MBOL framework for multicarrier ISAC systems where the proposed method allocates power and subcarrier resources in a shared spectrum scenario.

The effectiveness of the proposed MBOL method is compared to the Model-Free Online Learning (MFOL) method and other baseline methods in numerical experiments. The empirical results demonstrate that the MBOL method is highly effective in different interference environments. The MBOL method is more data-efficient and has the benefits of a model-based approach, including explainability, the ability to exploit useful information, and the proven convergence to the optimal policy under certain assumptions. However, when a large number of training samples are available or when the assumptions of the MBOL approach are not met, the MFOL method may achieve better performance than the MBOL methods. The results also show that the derived MBOL bounds are not tight for the resource allocation problem considered in this paper. Thus, considerably better convergence rates are achieved in practice. We numerically investigate a specific lemma in the proof and show that it is the root cause for the lack of tightness.

Future research may focus on extending our method to handle reactive interference better. This is particularly important for security and defense applications, where integrating these effects into controller design and model learning may significantly enhance system performance.

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Appendix A
Proofs of Theorems 2 and 3

The proof of Theorems 2 and 3 uses two lemmas provided with proofs in this appendix. The first lemma states the following.

Lemma 1. *Under Assumption 6 defined in Section VI-A1, the difference between the true expected utility and the model-based utility is bounded above by*

$$|u_n(\mathbf{s},\mathbf{a}) - u_n^{\theta}(\mathbf{s},\mathbf{a})| \leq u_{\max}\sqrt{2\mathcal{D}_{\mathrm{KL}}\left(\theta,\mathbf{s}\right)}. \tag{37}$$

where $u_{\max}$ is the maximum utility, and $\mathcal{D}_{\mathrm{KL}}\left(\theta,\mathbf{s}\right)$ is the Kullback–Leibler (KL) divergence between the true distribution and the model distribution conditioned on the current state as defined in (10).

Proof. We can write that

$$\begin{aligned}
&|u_n(\mathbf{s},\mathbf{a}) - u_n^{\theta}(\mathbf{s},\mathbf{a})| && (38)\\
&= \left|\int_{\mathbf{s}'\in\mathcal{S}} \left[\mathrm{T}(\mathbf{s}'|\mathbf{s}) - \mathrm{T}_{\theta}(\mathbf{s}'|\mathbf{s})\right] u_n(\mathbf{s},\mathbf{s}',\mathbf{a}) d\mathbf{s}'\right| \\
&\leq \int_{\mathbf{s}'\in\mathcal{S}} \left|\mathrm{T}(\mathbf{s}'|\mathbf{s}) - \mathrm{T}_{\theta}(\mathbf{s}'|\mathbf{s})\right| u_n(\mathbf{s},\mathbf{s}',\mathbf{a}) d\mathbf{s}' && (39)\\
&\leq 2u_{\max}\mathcal{D}_{\mathrm{TV}}\left(\mathrm{T}(\cdot|\mathbf{s}), \mathrm{T}_{\theta}(\cdot|s)\right) && (40)
\end{aligned}$$

where $\mathbf{s}'$ indicates the next state, and the definition of total variation distance (TVD) is used to obtain (40). Then we complete the proof by applying Pinsker's inequality

$$\mathcal{D}_{\mathrm{TV}}\left(\mathrm{T}(\cdot|\mathbf{s}), \mathrm{T}_{\theta}(\cdot|s)\right) \leq \sqrt{\frac{1}{2}\mathcal{D}_{\mathrm{KL}}\left(\theta,\mathbf{s}\right)}. \tag{41}$$

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□

Lemma 2. *In unconstrained setting and under Assumption 6, the gap between the expected utility of an optimal action* $\mathbf{a}^* = \pi^*(\mathbf{s})$ *and model-based agent action* $\hat{\mathbf{a}} = \pi_\theta(\mathbf{s})$ *is bounded above by*

$$|u_0(\mathbf{s}, \mathbf{a}^*) - u_0(\mathbf{s}, \hat{\mathbf{a}})| \leq 2\sqrt{2} u_{\max} \sqrt{\mathcal{D}_{\mathrm{KL}}(\theta, \mathbf{s})}. \tag{42}$$

Proof. In the unconstrained setting and under Assumption 1, it is easy to see that the optimal policy $\forall \mathbf{s} \in \mathcal{S}$ is

$$\pi^*(\mathbf{s}) = \arg\max_{\mathbf{a} \in \mathcal{A}} u_0(\mathbf{s}, \mathbf{a}). \tag{43}$$

In this proof, we utilize Lemma 1 and the following shorthand notation

$$\epsilon_\theta(\mathbf{s}) := u_{\max} \sqrt{2\mathcal{D}_{\mathrm{KL}}(\theta, \mathbf{s})}. \tag{44}$$

Thus we can write that

$$\begin{aligned} & u_0(\mathbf{s}, \mathbf{a}^*) - u_0(\mathbf{s}, \hat{\mathbf{a}}) && (45)\\ & \leq u_0^\theta(\mathbf{s}, \mathbf{a}^*) + \epsilon_\theta(\mathbf{s}) - u_0(\mathbf{s}, \hat{\mathbf{a}}) \\ & \leq u_0^\theta(\mathbf{s}, \hat{\mathbf{a}}) - u_0(\mathbf{s}, \hat{\mathbf{a}}) + \epsilon_\theta(\mathbf{s}) && (46)\\ & \leq 2\epsilon_\theta(\mathbf{s}) && (47) \end{aligned}$$

Using the definition in (44) completes the proof. □

Finally, Theorem 2 introduced in Section VI-A2 is proven by using Lemma 2. Thus, we obtain that

$$\begin{aligned} & J_0(\pi^*) - \mathbb{E}\left[J_0(\pi_K)\right] && (48)\\ & = \mathbb{E}\left[\mathbb{E}_\mu\left[u_0(\mathbf{s}, \pi^*(\mathbf{s})) - u_0(\mathbf{s}, \pi_{\bar{\theta}_K}(\mathbf{s}))|\bar{\theta}_K\right]\right] \\ & \leq 2\sqrt{2} u_{\max} \mathbb{E}\left[\mathbb{E}_\mu\left[\sqrt{\mathcal{D}_{\mathrm{KL}}(\bar{\theta}_K, \mathbf{s})}\,\middle|\,\bar{\theta}_K\right]\right] && (49)\\ & \leq 2\sqrt{2} u_{\max} \sqrt{\mathbb{E}\left[L(\bar{\theta}_K)\right]}. && (50) \end{aligned}$$

where in (50) we use Jensen inequality and the definition in (11). We complete the proof of Theorem 2 by applying Theorem 1 provided in Section VI-A1 to (50).

Furthermore, Theorem 3 in Section VI-A3 is proven by using Lemma 1 and Assumption 7 such that for all $n = 1, \ldots, N_C$ we have that

$$\begin{aligned} J_n(\pi_\theta) &= \mathbb{E}_\mu\left[u_n(\mathbf{s}, \pi_K(\mathbf{a})\right] && (51)\\ &\geq \mathbb{E}_\mu\Big[\underbrace{u_n^\theta(\mathbf{s}, \pi_\theta(\mathbf{a}))}_{\geq C_n} - u_{\max}\sqrt{2\mathcal{D}_{\mathrm{KL}}(\theta, \mathbf{s})}\Big] && (52)\\ &\geq C_n - u_{\max}\sqrt{2L(\theta)}. && (53) \end{aligned}$$

where C_n is the lower-bound constraint. Applying Theorem 1 to

$$\mathbb{E}\left[J_n(\pi_K)\right] \geq C_n - u_{\max}\sqrt{2\mathbb{E}\left[L(\bar{\theta}_K)\right]} \tag{54}$$

completes the proof for Theorem 3.

Appendix B
Relation to Finite State Markov Chains

Consider the model proposed in Section V-C1. Without loss of generality, we can remove the indices for $x \in \{r, c\}$ and the parameters θ to simplify the notation. Therefore, we write that

$$\mathrm{T}(\mathbf{s}_{k+1}|\mathbf{s}_k) = \prod_{i=1}^{N}\prod_{j=1}^{L}\left[\frac{\phi_{i,j}(\mathbf{s}_k)}{\mathcal{V}_j}\right]^{\mathbb{1}_{\{\mathbf{s}_{k+1,i} \in \mathcal{C}_j\}}} \tag{55}$$

where N is the number of sub-channels, L is the number of approximation bins, and $\phi_{i,j}$ is the probability weight function. Define subsets $\mathcal{S}_l \subset \mathcal{S}$ for all $l = 1, \ldots, L^N$ as follows

$$\mathcal{S}_l = \{\mathbf{s} \in \mathcal{S} : s_i \in \mathcal{C}_{\mathcal{I}_l(i)}\ \forall\, i = 1, \ldots, N\} \tag{56}$$

where $\mathcal{I}_l(i)$ is an index function that covers all combinations of channel bins. Using (56), we can write the transition distribution as follows

$$\begin{aligned} & \mathrm{T}(\mathbf{s}_{k+1}|\mathbf{s}_k) && (57)\\ &= \sum_{l=1}^{L^N} \mathbb{1}_{\{\mathbf{s}_{k+1} \in \mathcal{S}_l\}}\left(\prod_{i=1}^{N}\prod_{j=1}^{L}\left[\frac{\phi_{i,j}(\mathbf{s}_k)}{\mathcal{V}_j}\right]^{\mathbb{1}_{\{\mathbf{s}_{k+1,i} \in \mathcal{C}_j\}}}\right) \\ &= \sum_{l=1}^{L^N} \mathbb{1}_{\{\mathbf{s}_{k+1} \in \mathcal{S}_l\}} \prod_{i=1}^{N} \frac{\phi_{i,\mathcal{I}_l(i)}(\mathbf{s}_k)}{\mathcal{V}_{\mathcal{I}_l(i)}} && (58)\\ &= \sum_{k=1}^{L^N} \mathbb{1}_{\{\mathbf{s}_{k+1} \in \mathcal{S}_l\}} \frac{\widetilde{\phi}_l(\mathbf{s}_k)}{\widetilde{\mathcal{V}}_l} && (59) \end{aligned}$$

where $\widetilde{\phi}_l(\mathbf{s}_k) := \prod_{i=1}^{N} \phi_{i,\mathcal{I}_l(i)}(\mathbf{s}_k)$ and $\widetilde{\mathcal{V}}_l := \prod_{i=1}^{N} \mathcal{V}_{\mathcal{I}_l(i)}$. We use notation $\mathrm{T}^K(\cdot|\cdot) = \Pr\{S_{t+K}|S_t\}$ to denote $K \in \mathbb{N}$ step transition probabilities, and we have that $T^1 = T$. The $K = 2$ step transition function from state $\mathbf{s}$ to $\mathbf{s}''$ is

$$\begin{aligned} T^2(\mathbf{s}''|\mathbf{s}) &= \int_{\mathbf{s}' \in \mathcal{S}} T(\mathbf{s}''|\mathbf{s}')T(\mathbf{s}'|\mathbf{s})d\mathbf{s}' && (60)\\ &= \sum_{l=1}^{L^N} \int_{\mathbf{s}' \in \mathcal{S}_l} T(\mathbf{s}''|\mathbf{s}')T(\mathbf{s}'|\mathbf{s})d\mathbf{s}' && (61)\\ &= \sum_{l=1}^{L^N} \int_{\mathbf{s}' \in \mathcal{S}_l} \Bigg(\sum_{n=1}^{L^N} \mathbb{1}_{\{\mathbf{s}'' \in \mathcal{S}_n\}} \frac{\widetilde{\phi}_n(\mathbf{s}')}{\widetilde{\mathcal{V}}_n} && (62)\\ &\qquad \times \sum_{m=1}^{L^N} \mathbb{1}_{\{\mathbf{s}' \in \mathcal{S}_m\}} \frac{\widetilde{\phi}_m(\mathbf{s})}{\widetilde{\mathcal{V}}_m}\Bigg) d\mathbf{s}' \\ &= \sum_{l=1}^{L^N}\sum_{n=1}^{L^N} \mathbb{1}_{\{\mathbf{s}'' \in \mathcal{S}_n\}} \frac{\widetilde{\phi}_l(\mathbf{s})}{\widetilde{\mathcal{V}}_n} \frac{1}{\widetilde{\mathcal{V}}_l} \int_{\mathbf{s}' \in \mathcal{S}_l} \widetilde{\phi}_n(\mathbf{s}')d\mathbf{s}' && (63)\\ &= \sum_{n=1}^{L^N} \frac{\mathbb{1}_{\{\mathbf{s}'' \in \mathcal{S}_n\}}}{\widetilde{\mathcal{V}}_n} \sum_{l=1}^{L^N} \widetilde{\phi}_l(\mathbf{s})\bar{\phi}_{l,n} && (64)\\ &= \sum_{n=1}^{L^N} \frac{\mathbb{1}_{\{\mathbf{s}'' \in \mathcal{S}_n\}}}{\widetilde{\mathcal{V}}_n} \left[\mathbf{\Phi}^{\mathrm{T}} \widetilde{\boldsymbol{\phi}}(\mathbf{s})\right]_n && (65) \end{aligned}$$

where

$$\bar{\phi}_{l,n} := \frac{1}{\widetilde{\mathcal{V}}_l} \int_{\mathbf{s}' \in \mathcal{S}_l} \widetilde{\phi}_n(\mathbf{s}')d\mathbf{s}' \tag{66}$$

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This article has been accepted for publication in IEEE Journal of Selected Topics in Signal Processing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JSTSP.2024.3386057

is the average transition probability from $\mathcal{S}_l$ to bin $\mathcal{S}_n$. Furthermore, elements of the matrix $\mathbf{\Phi}$ and the vector $\widetilde{\boldsymbol{\phi}}(\mathbf{s})$ correspond to $\bar{\phi}_{l,n}$ and $\widetilde{\phi}_l(\mathbf{s})$, respectively. From (65), it is straightforward to generalize for all $K \in \mathbb{N}$ by induction

$$T^K(\mathbf{s}'|\mathbf{s}) = \sum_{n=1}^{L^N} \frac{\mathbb{1}_{\{\mathbf{s}' \in \mathcal{S}_n\}}}{\widetilde{\mathcal{V}}_n} \left[\left(\mathbf{\Phi}^{K-1}\right)^{\mathrm{T}} \widetilde{\boldsymbol{\phi}}(\mathbf{s}) \right]_n . \tag{67}$$

Thus, (67) shows that the predictive state transitions for $K > 1$ are generated by finite state Markov chain with a transition probability matrix $\mathbf{\Phi}$.

Appendix C
Model learning regret and the problem dependent constants

This appendix studies the model learning problem for the linear MBOL method considered in Section VII. The method uses the Online Gradient Descent (OGD) learning method, the probability weights defined in (33), and the channel-wise one-hot encoded mapping defined in (35).

The regret analysis in [15] can be used to obtain the following regret bound for the OGD algorithm

$$\mathcal{R}_K^{\mathcal{M}} = \frac{3}{2} G D \sqrt{K} \tag{68}$$

where K is the number of learning steps, G is the Lipschitz constant, and D is the parameter space dimension. The instantaneous loss functions in Section V-C1 are defined as follows

$$l(\theta; \mathbf{s}_k, \mathbf{s}_{k+1}) = \xi - \sum_{i=1}^{N} \sum_{j=1}^{L} \mathbb{1}_{\{s_{k+1,i} \in \mathcal{C}_j\}} \log \phi_{i,j}(\mathbf{s}_k, \theta). \tag{69}$$

Without loss of generality, we can remove the indices $x \in \{r, c\}$ to simplify the notation. Let us define a short-hand notation $l_k(\theta) \coloneqq l(\theta; \mathbf{s}_k, \mathbf{s}_{k+1})$. Next, we derive G and D to show how the regret scales with respect to the system parameters, such as the number of channels N and approximation bins L. The G bounds the L_2-norm of the gradient $\nabla_\theta l_k(\theta)$ from above. For any $i \in \{1, \ldots, N\}$ and $j \in \{1, \ldots, L\}$, the gradient can be written as follows

$$\nabla_{\theta_{i,j}} l_k(\theta) = [\phi_{i,j}(\mathbf{s}_k, \theta) - \mathbb{1}_{\{s_{k+1,i} \in \mathcal{C}_j\}}] \boldsymbol{\psi}(\mathbf{s}). \tag{70}$$

Note that the dimension of the mapping $\boldsymbol{\psi}(\mathbf{s})$ is $M = NL$, but there are only N non-zero elements with values 1. Thus we can obtain the bound

$$\|\nabla_\theta l_k(\theta)\|_2 = \sqrt{\sum_{i=1}^{N} \sum_{j=1}^{L} \sum_{l=1}^{M} ([\nabla_{\theta_{i,j}} l_{k+1}(\theta)]_l)^2} \tag{71}$$

$$\leq N\sqrt{L} \tag{72}$$

Similarly, we can bound $\|\theta - \theta^*\|_2 \leq \rho N L$ where $\|\theta - \theta^*\|_\infty \leq \rho$ for any θ and θ^* in Θ. Using these definitions, the model learning regret bound can be written as follows

$$\mathcal{R}_K^{\mathcal{M}} = \frac{3}{2} N^2 L^{\frac{3}{2}} \rho \sqrt{K}. \tag{73}$$

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